



Standard Test Method for Corrosive Sulfur in Electrical Insulating Oils¹

This standard is issued under the fixed designation D 1275; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been adopted for use by government agencies to replace Method 5328-2 of Federal Test Method Standard No. 791b. This standard was adopted as an ASTM-IP Standard.

1. Scope

1.1 This test method describes the detection of corrosive sulfur compounds (both inorganic and organic) in electrical insulating oils of petroleum origin.

1.2 New and in-service mineral insulating oils may contain substances that cause corrosion under certain conditions of use. This test method is designed to detect the presence of, or the propensity to form, free (elemental) sulfur and corrosive sulfur compounds by subjecting copper to contact with oil under prescribed conditions.

1.3 Two methods (A and B) have been introduced where historically there has only been one. Method A is the one that has been in place since 1953. Method B is more rigorous and the preferred method.

1.4 The values stated in SI units are to be regarded as the standard. Inch-pound units are included for informational purposes.

1.5 *This standard does not purport to address the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.*

2. Referenced Documents

2.1 *ASTM Standards:*²

D 130 Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

2.2 *Other Document:*

ANSI B74.10 Grading of Abrasive Microgrits³

¹ This test method is under the jurisdiction of ASTM Committee D27 on Electrical Insulating Liquids and Gases and is the direct responsibility of Subcommittee D27.06 on Chemical Test.

Current edition approved March 1, 2006. Published March 2006. Originally approved in 1953. Last previous edition approved in 2003 as D 1275 – 03.

² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

³ Available from American National Standards Institute, 25 W. 43rd St., 4th Floor, New York, NY 10036, USA.

3. Summary of Test Methods

3.1 *Method A*—250 mL of oil is aged in a sealed flask for 19 h at 140°C in the presence of a copper strip.

3.2 *Method B*—220 mL of oil is aged in a sealed heavy-walled bottle for 48 h at 150°C in the presence of a copper strip. This is the preferred method.

4. Significance and Use

4.1 In most of their uses, insulating oils are continually in contact with metals that are subject to corrosion. The presence of corrosive sulfur compounds will result in deterioration of these metals. The extent of deterioration is dependent upon the quantity and type of corrosive agent and time and temperature factors. Detection of these undesirable impurities, even though not in terms of quantitative values, is a means for recognizing the hazard involved.

5. Apparatus

5.1 *Bath*—A hot-air oven or oil bath provided with suitable means of heating to, and controlling at 140 or 150 ± 2°C. A circulating hot-air oven is preferred.

5.2 *Flasks*—Narrow-mouth, 250-mL, ground-glass stoppered flasks, of chemically resistant glass, capable of holding 270 to 280 mL when filled completely to the stopper. Flasks of such capacity are required in order to allow sufficient space for expansion of the oil.

5.3 *Bottles*⁴—Heavy walled, 250 mL, bottles of chemically resistant glass constructed with necks to receive a PTFE threaded plug equipped with a fluoro-elastomer o-ring. Bottles of such capacity and design are required in order to allow sufficient space for expansion of the oil and eliminate intrusion from atmospheric gases.

5.4 *Copper Foil, 99.9+ % pure, 0.127 to 0.254 mm (0.005 to 0.010 in.) in thickness.*

⁴ The sole source of supply of the bottles and PTFE screw plugs known to the committee at this time is Ace Glass, P.O. Box 688, Vineland, NJ 08362, USA. If you are aware of alternative suppliers, please provide this information to ASTM International Headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee,¹ which you may attend.

5.5 *Polishing Material*, consisting of 240-grit silicon carbide paper or cloth, and also 230-mesh silicon carbide grains and pharmaceutical absorbent cotton.

NOTE 1—It should be noted that 240-grit silicon carbide paper and 230-mesh silicon carbide grains have particle sizes of about the same size (63 μm). In the United States, abrasive papers are classified in accordance with ANSI B74.10. Abrasive powders are classified by ASTM mesh size.

6. Reagents

6.1 *Acetone*, cp.

6.2 *Nitrogen Gas*—Commercial cylinders of nitrogen gas are satisfactory for this purpose.

7. Preparation of Apparatus

7.1 Chemically clean flasks, glass stoppers, bottles and PTFE screw plugs with solvents to remove oil, then wash the flasks with sulfur-free cleaning powder or liquid. Rinse with tap water, then with distilled water, and dry in an oven.

7.2 Cut a strip of copper 6 by 25 mm ($\frac{1}{4}$ by 1 in.) (Note 2) and remove blemishes from surfaces with the 240-grit silicon carbide paper. Strips may be stored in sulfur-free acetone at this point for future use. Do the final polishing of the strip by removing it from the acetone, holding it in the fingers protected with ashless filter paper or nitrile gloves, and rubbing with 230-mesh silicon carbide grains picked up from a glass plate with a pad of absorbent cotton moistened with a drop of acetone. Wipe the strip with fresh pads of cotton and subsequently handle only with stainless steel forceps (do not touch with the fingers). Rub in the direction of the long axis of the strip. Clean all metal dust and abrasive from the strip, using successive clean cotton pads until a fresh pad remains unsoiled. Bend the clean strip in a V-shape at approximately a 60° angle and wash successively in acetone, distilled water, and acetone. Dry in an oven for 3 to 5 min at 80 to 100°C and immediately immerse the copper strip in the prepared test specimen of oil (Note 3). Do not use compressed air or an inert gas to dry the copper strip.

NOTE 2—It has been found convenient to polish a larger piece of copper from which, after the final polishing, several strips of the proper size may be cut.

NOTE 3—This method of cleaning has been adapted from Test Method D 130.

8. Method A

8.1 Use the oil to be tested as received. Do not filter the oil through paper.

8.2 Promptly place the prepared copper strip in a clean 250-mL flask to which has been added 250 mL of the oil to be tested. Place the bent copper strip standing on its long edge so that no flat surface lies along the glass bottom of the vessel. Lubricate the ground-glass stopper with a small amount of the test specimen. Bubble nitrogen through the oil in the flask by means of a glass tube connected to the reduction or needle valve of the cylinder (rubber connections must be sulfur-free) for 1 min, and quickly put the stopper loosely in place.

8.3 Place the stoppered flask (immersed to the neck in the event an oil bath is employed) in the oven at 140°C . When the oil in the flask has reached approximately 140°C , tighten the stopper more firmly (Note 4). Remove the flask after heating

for $19\text{ h} \pm 10\text{ min}$ at $140 \pm 2^\circ\text{C}$. Carefully take the copper strip from the flask and wash with acetone or other suitable solvent to remove all of the oil and let air dry. Do not use pressurized air to dry copper strip.

NOTE 4—The corrosive effects of unstable sulfur compounds present in the oil are reduced when aged under strong oxidizing conditions such as when oxygen is present. It is imperative to keep the flask well sealed after nitrogen bubbling. If the oil visibly darkens during testing then most likely oxygen has entered the vessel and oxidized the oil. In these cases, the test procedure has become compromised and must be repeated.

8.4 To inspect, hold the test strip in such a manner that light reflected from it at an angle of approximately 45° will be observed.

9. Method B

9.1 Prepare the test specimen as described in 8.1.

9.2 Promptly place the prepared copper strip in a clean 250-mL bottle to which has been added 220 mL of the oil to be tested. Place the bent copper strip standing on its long edge so that no flat surface lies along the glass bottom of the vessel. Bubble nitrogen through the oil in the bottle by means of a $\frac{1}{16}$ in. inner diameter glass or stainless steel tube connected to the reduction or needle valve of the cylinder (rubber connections must be sulfur-free) for 5 min at a rate of 0.5 L/min. Quickly screw in the PTFE plug equipped with fluoro-elastomer o-ring.

9.3 Place the bottle in the oven at 150°C . After 15 min of heating at 150°C , partially unscrew the PTFE plug to release the pressure and then screw it back down so the bottle does not burst. Remove the bottle after heating for $48\text{ h} \pm 20\text{ min}$ at $150 \pm 2^\circ\text{C}$. Allow to cool. Carefully take the copper strip from the flask and wash with acetone or other suitable solvent to remove all of the oil and let air dry. Do not use pressurized air to dry copper strip.

9.4 To inspect, hold the test strip in such a manner that light reflected from it at an angle of approximately 45° will be observed. If a specimen is borderline or unclear, take a clean laboratory tissue and drag it across the surface with heavy pressure; if any of the deposited material is removed, the deposit is corrosive and is to be reported as such.

10. Interpretation of Results

10.1 Classify the oil as corrosive or noncorrosive in accordance with Table 1. Classification of corrosive or noncorrosive is to be aided through the use of ASTM Copper Strip Corrosion Standards as referenced in Test Method D 130. A description of the tarnish levels is provided in Table 2 for reference only.

11. Report

11.1 Report the following information:

11.1.1 Sample identification, and

TABLE 1 Copper Strip Classifications

Classification	Description
Noncorrosive	Orange, red, lavender, multicolored with lavender blue or silver, or both, overlaid on claret red, silvery, brassy or gold, magenta overcast on brassy strip, multicolored with red and green showing (peacock) but no gray
Corrosive	Transparent black, dark gray or dark brown, graphite or lusterless black, glossy or jet black, any degree of flaking

**TABLE 2 Copper Strip Tarnish Level Classifications**

Classification	Designation	Description
Freshly Polished Strip	...	Unable to reproduce upon aging so no description is provided
1	Slight tarnish	a. Light orange, almost the same as freshly polished strip b. Dark orange
2	Moderate tarnish	a. Claret red b. Lavender c. Multicolored with lavender blue or silver, or both, overlaid on claret red d. Silvery e. Brassy or gold
3	Dark tarnish	a. Magenta overcast on brassy strip b. Multicolored with red and green showing (peacock), but no gray
4	Corrosion	a. Transparent black, dark gray or brown with peacock green barely showing b. Graphite or lusterless black c. Glossy or jet black

11.1.2 Test specimen as being corrosive or noncorrosive,

11.1.3 Tarnish level according to Test Method **D 130**, and

11.1.4 Method A or Method B.

12. Precision and Bias

12.1 No statement is made about either the precision or bias of this test method since the result merely states whether there is conformance to the criteria for success specified in the procedure and is not quantitative.

13. Keywords

13.1 corrosion; corrosive sulfur; elemental sulfur; free sulfur; insulating; mineral; oils; petroleum; sulfur

ASTM International takes no position respecting the validity of any patent rights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of the validity of any such patent rights, and the risk of infringement of such rights, are entirely their own responsibility.

This standard is subject to revision at any time by the responsible technical committee and must be reviewed every five years and if not revised, either reapproved or withdrawn. Your comments are invited either for revision of this standard or for additional standards and should be addressed to ASTM International Headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee, which you may attend. If you feel that your comments have not received a fair hearing you should make your views known to the ASTM Committee on Standards, at the address shown below.

This standard is copyrighted by ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, United States. Individual reprints (single or multiple copies) of this standard may be obtained by contacting ASTM at the above address or at 610-832-9585 (phone), 610-832-9555 (fax), or service@astm.org (e-mail); or through the ASTM website (www.astm.org).